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Stem-cell-based human and mouse embryo models

Min Bao[1,2,#,*], Jake Cornwall-Scoones[1,3,#,*] and Magdalena Zernicka-Goetz[1,2,*]

Synthetic embryology aims to develop embryo-like structures from stem cells to provide new insight into early stages of mammalian development. Recent advances in synthetic embryology have highlighted the remarkable capacity of stem cells to self-organize under certain biochemical or biophysical stimulations, generating structures that recapitulate the fate and form of early mouse/human embryos, in which symmetry breaking, pattern formation, or proper morphogenesis can be observed spontaneously. Here we review recent progress on the design principles for different types of embryoids and discuss the impact of different biochemical and biophysical factors on the process of stem-cell self-organization. We also offer our thoughts about the principal future challenges.

Addresses
[1] California Institute of Technology, Division of Biology and Biological Engineering, 1200 E. California Boulevard, Pasadena, CA 91125, USA
[2] Mammalian Embryo and Stem Cell Group, Department of Physiology, Development, and Neuroscience, University of Cambridge, Cambridge CB2 3DY, UK
[3] The Francis Crick Institute, London NW1 1AT, UK

Corresponding author: Magdalena Zernicka-Goetz (mz205@cam.ac.uk)
[*] Twitter account: @Min_Bao_, @jake_cs_, @ZernickaGoetz
[#] These authors contributed equally

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Introduction

During the early mammalian development, embryos undergo profound morphological and cell-state transitions to establish distinctive architectures and axially patterned cell-type diversity before gastrulation [1–3]. Despite their importance, underpinning mechanisms often remain undefined. Principal challenges in elucidating these processes include their multi-scale nature, their spatiotemporal control, and their 'emergent' character.

Synthetic embryology has evolved as an interdisciplinary field to tackle these challenges, combining principles from stem-cell biology, developmental biology, and the engineering sciences with the goal of developing functional substitutes for mammalian embryos [4,5]. Recently, many key developmental events have been recapitulated *in vitro*, providing a better understanding of molecular mechanisms that coordinate lineage specification, developmental plasticity, and tissue morphogenesis across gastrulation. By exploiting the remarkable capacity of stem cells to self-organize into functional structures, these studies highlight the power of stem-cell-derived embryo models ('embryoids') in dissecting the physical basis of embryonic morphogenesis [2,6–9].

In this review, we introduce the major embryo models, which we call embryoids, highlighting the importance of different biophysical and biochemical factors for their self-organization. Viewing *in vitro* reconstruction as an engineering problem, we define the components necessary for the generation of embryoids: the cellular constituents, the chemical environment, and the physical context. We propose that different approaches strike a balance between precise extrinsic control and a reliance on the intrinsic self-organizing capacity of stem cells. We also discuss discoveries already made with these systems, and outline the challenges and prospects of synthetic embryology in the future.

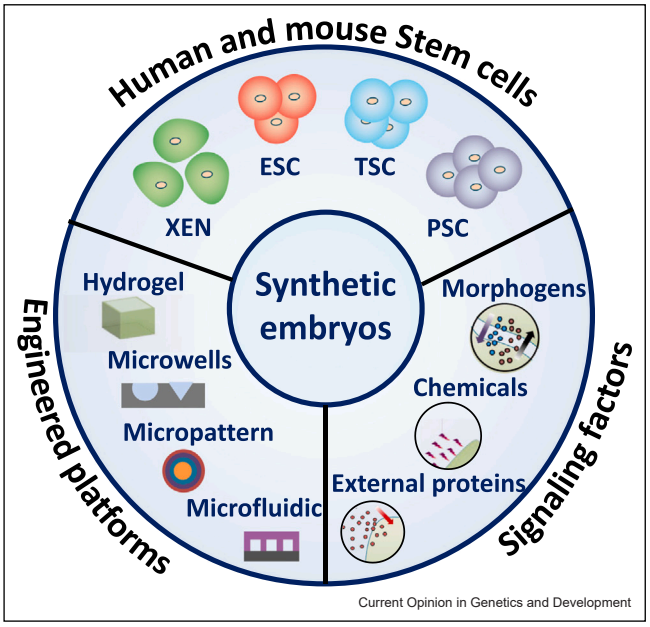
Synthetic embryology strategies: alternate approaches for a common goal

Synthetic embryology attempts to recapitulate aspects of early development using stem cells cultured *in vitro*. Approaches differ depending on the species and developmental stage, as well as the aspect of development under investigation. By providing alternative and complementary perspectives, embryoids are particularly well-suited to decipher principles of self-organization. Instead of providing a comprehensive overview of all the present strategies (we refer readers to excellent reviews [2,6,10–13]), below, we consider the three major methodological components required to build embryo-like structures *in vitro*: (i) the types of stem cell used; (ii) the biochemical conditions; and (iii) the physical environment (Figure 1).

The raw material: which stem cells?

Early mammalian development involves the coordination of cell-fate specification and morphogenesis.

Figure 1



Strategies for synthetic embryology. Three major compartments contribute toward the creation of an embryo-like structure: stem cells, signaling factors, and engineered platforms. PSC, pluripotent stem cell.

Starting from the totipotent zygote, cells undergo sequential fate bifurcations, establishing three principal lineages by the time of implantation: epiblast, trophectoderm, and primitive endoderm. At gastrulation, the epiblast breaks symmetry to lay down the embryonic axes, differentiating into the three germ layers (endoderm, mesoderm, and ectoderm). Further, throughout development, the epiblast undergoes multiple transcriptional transformations, notably in the naive-to-primed transition [14–16]. Stem cell-derived embryo models must thus consider which raw material to utilize, and in which combination. Given the multilineage composition of the embryo, with patterning and morphogenesis relying on intertissue interactions, different approaches strike a balance between recapitulating these interactions or relying on the properties and autonomous self-organizing capacity of embryonic stem cells (ESCs).

Many approaches in synthetic embryology rely solely on ESCs, which resemble the epiblast both transcriptionally and morphologically. When cultured under appropriate conditions, ESCs can undergo embryo-like patterning. Early successes in synthetic embryology centered upon aggregates of mouse ESCs — termed embryoid bodies (EBs), which are able to develop to specify mesoderm when BMP or Wnt pathways are stimulated [17]. 3D gastruloids extend this EB model by pulsing aggregates of ESCs with the Wnt agonist CHIRON [18]. These structures recapitulate aspects of gastrulation-stage patterning, undergoing axial elongation and displaying molecular signatures of the anterior–posterior and dorsoventral axes [18–20••]. However, they do not undertake a principal hallmark of mammalian gastrulation, the epithelial-to-mesenchymal (EMT) transition, and associated cell migration. The self-organizing capacity of ESCs appears conserved: human ESC (hESC) aggregates pulsed with CHIRON also generate gastruloids [21]. The principles of human gastrulation-stage cell-fate specification have however been best defined in 2D micropatterned hESC colonies. Spatially confining hESCs onto circular micropatterns promotes patterned differentiation upon BMP4 stimulation, with germ-layer markers being expressed in concentric rings: ectoderm centrally, extraembryonic tissue peripherally, and with mesoderm/endoderm in-between [22••]. Alternatively, aspects of gastrulation- and neurulation-stage morphogenesis have been modeled in human amniotic sac embryoids [23].

Beyond successes in exploiting the autonomous self-organizing capacity of ESCs, alternative strategies have relied on the fostering self-assembly of multiple stem-cell types, taking into account the essential role for embryonic/extraembryonic cross-talk in natural development [11,24,25•]. Mouse blastocyst-like structures, blastoids, can be generated by culturing ESCs with trophoblast stem cells (TSCs), mimicking the trophectoderm [26••]. While recapitulating blastocyst morphology, early blastoid protocols demonstrated poor primitive endoderm lineage commitment. Blastoids

generated using extended potential ESCs in combination with TSCs developed all three blastocyst lineages [27,28], highlighting the importance of using appropriate starting materials. This approach has recently been adapted to recapitulate the human blastocyst [29••–32]. Human blastoids resemble blastocysts in morphology and gene expression, and could initiate development of peri-implantation-like structures [31••,33], providing a way to investigate aspects of human embryogenesis *in vitro*.

Combining different types of stem cells has also proved successful in modeling postimplantation development. This was first shown by our lab, when ESCs and TSCs were aggregated in Matrigel to generate so-called ETS embryos, recapitulating both proamniotic cavity formation and anterior–posterior symmetry breaking, leading to formation of primordial germ cells, mesoderm and endoderm [34••]. To build a complete embryoid, with all its distinct lineages and compartments, our lab subsequently established conditions to foster assembly of not only ESCs and TSCs with each other, but also with the other extra-ESC, extraembryonic endoderm (XEN) stem cells, representing the visceral endoderm [35••]. XEN cells secrete the extracellular matrix to establish a basement membrane (BM), enveloping both ESCs and TSCs. From 4 days after cell seeding, these complete embryoids have the capacity to undertake EMT, gastrulation, and germ-layer specification [35••]. To enhance the efficiency of gastrulation, we recently replaced XEN cells with ESCs that transiently express the primitive endoderm master transcription factor Gata4 in response to doxycycline to generate induced embryoids. In addition to expressing canonical postimplantation markers in all three compartments, induced embryoids recapitulate key morphogenetic events, leading to migration of the anterior signaling center and fully accomplished gastrulation. Thus, replacement of XEN cells led to a new model of gastrulating embryoids [36•].

The chemical environment: which signaling factors?

Cell-fate specification in embryo relies largely on chemical cross-talk within and between lineages. These cell-fate decisions are coordinated in space by establishing and interpreting gradients of numerous diffusible signaling proteins (morphogens) [37], where alternate concentrations or combinations can elicit differential fate specification [38,39].

Roles for signaling interplay have been well-documented across mouse development. For example, studies of the epiblast-primitive endoderm fate bifurcation have highlighted roles for BMP and FGF in establishing the correct proportions of each lineage [40–44]. Morphogen cross-talk is seen perhaps most strikingly during gastrulation where the intersection of the BMP, Nodal, and Wnt pathways promotes germ-layer specification in the epiblast. Studies in the mouse have highlighted contributions from multiple tissues in establishing and stabilizing morphogen gradients: the secretion of BMP from the (trophectoderm-derived) extraembryonic ectoderm; and the role of Wnt and Nodal antagonists supplied by the (primitive endoderm-derived) anterior visceral endoderm, restricting the signaling activity to the posterior [12,44–46].

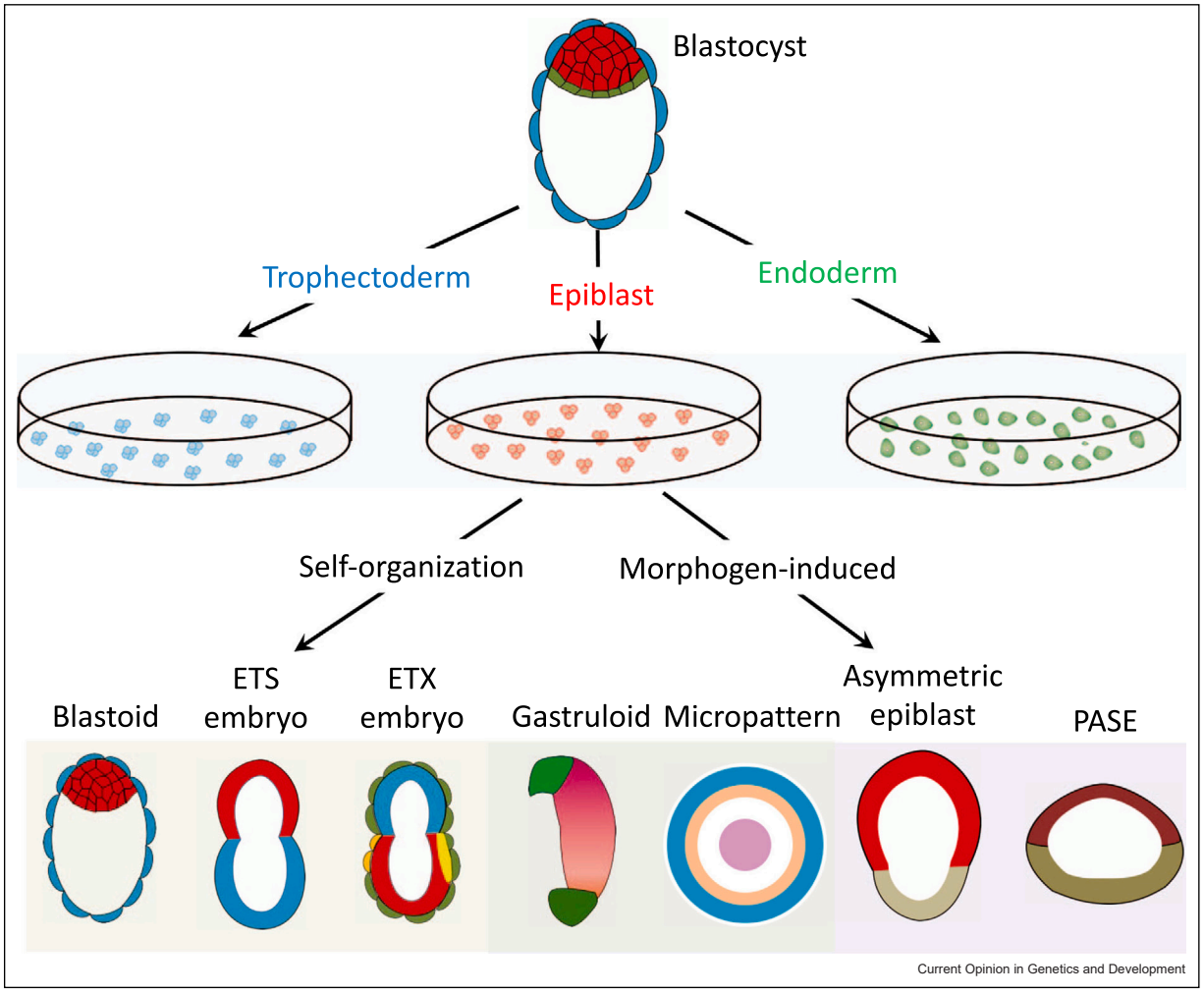
Given the influence of a cell's signaling environment on its fate *in vivo*, fate transitions can be induced *in vitro* by supplying the appropriate chemical cues to the culture medium. For example, XEN cells differentiate toward extraembryonic visceral endoderm fate in the presence of BMP4 and revert back after BMP4 removal [47]. Furthermore, Nodal-treated XEN cells differentiate toward embryonic visceral endoderm cells and activate transcriptomic signatures of the anterior visceral endoderm [48]. Fate transitions between the principal embryonic lineages can also be induced, given the appropriate signaling environment. Thus, for example, human TSCs can be generated from hESCs by culturing them with BMP4 and inhibitors of NODAL and FGF [49].

Approaches in synthetic embryology attempt to bring these concepts together, by striking a trade-off between tuning the culture conditions using knowledge of cell-fate control *in vitro*, and exploiting the capacity of cell communication to promote self-organization *in vivo* (Figure 2). Synthetic embryological approaches deploying ESCs alone typically stimulate patterning through a chemical cue: Wnt3a in EBs [17], Chiron in gastruloids [18,19,21], or Bmp4 in 2D ESC micropatterns [22••,50] and 3D ESC spheroids [51•]. Whereas the signaling molecules are provided homogeneously, the structures respond by self-organizing morphogen gradients, often via the expression of extracellular antagonists of signaling pathways [50,51•]. Inducing appropriate signaling can also trigger morphogenetic transitions, seen in the axial elongation of gastruloids spurred by Wnt-pathway activation [18,19], and in the role of cAMP and Wnt-pathway activators in promoting the blastoid cavitation [26••]. Indeed, this signaling cross-talk can be engineered: combining BMP4-treated mESC aggregates with untreated cells can promote a gastruloid-like structure, with BMP4 inducing the expression of Wnt3 and Nodal [52].

The physical context: which bioengineered platform?

Embryonic morphogenesis depends on the mechanical properties of cells and their environment. Indeed, *in vitro* studies have highlighted the importance of the

Figure 2



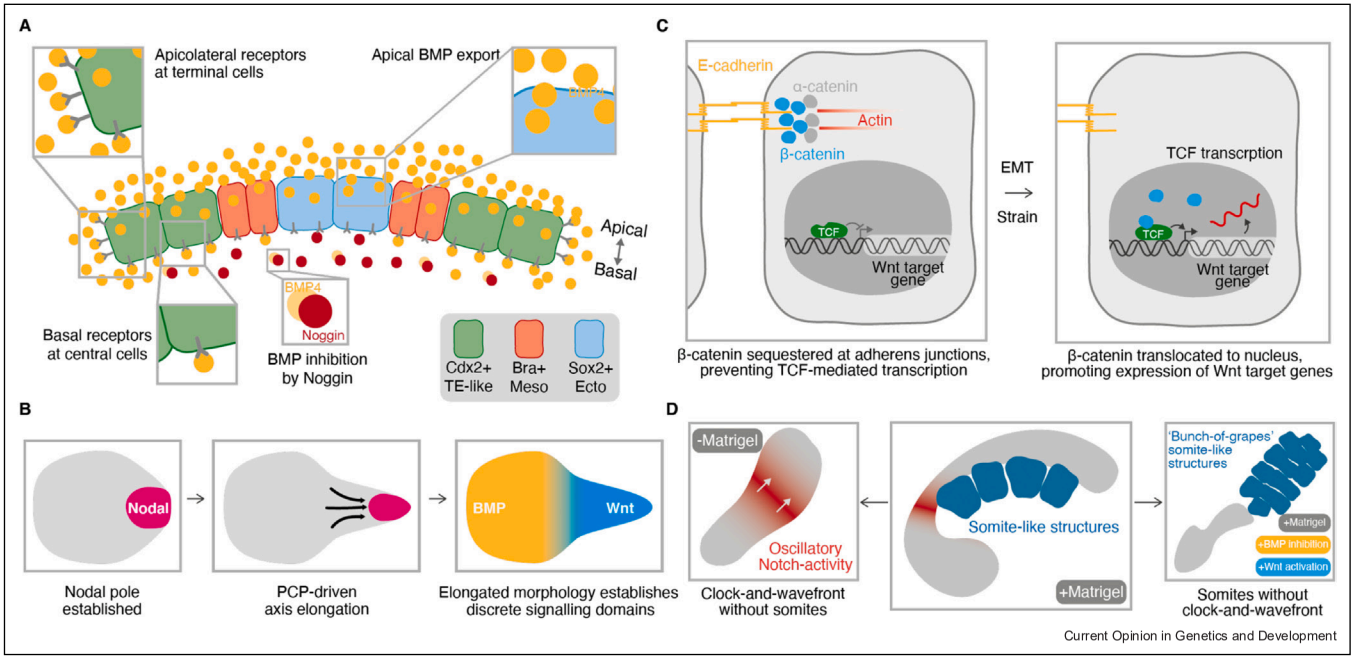
Modeling early mouse and human development with stem cells. Different stem cells that represent different lineages of an embryo can be derived from blastocyst and can be maintained *in vitro*. Self-organization-guided synthetic embryology is achieved by mixing different stem cells together, self-assembling and -patterning to establish embryo-like structures. Examples include blastoid, embryoids (ETS and ETX embryos). Morphogen-induced synthetic embryology involves pulsing stem-cell aggregates with exogenous chemical signals, facilitating cell-fate induction and morphogenesis. Examples include gastruloids, stimulated with CHIRON, and germ-layer micropatterns, treated with BMP4. (PASE, postimplantation amniotic sac embryoid, derived from primitive endoderm).

mechanical microenvironment on cell fate [53–56]. Consequently, advances in synthetic embryology have come hand-in-hand with technological refinements to the physical properties of the culture environment. Rapid development in material technology has provided new engineered culture platforms that contribute to stem-cell self-organization *in vitro*, including microwells, micropatterns, hydrogels, and microfluidic devices.

Many strategies for culturing embryoids rely on combining small numbers of stem cells. Microwell-based platforms have emerged as the principal solution, conducive to the accumulation of stem cells in appropriate numbers and proportions, enabling them to self-organize into blastocyst (blastoids [26••,28,57]) or gastrula-like structures (embryoids [35••,36•]). However, the present strategies remain imperfect. For example, the round or V-shape bottoms of commercially available microwells prohibit live imaging. Advances in bioengineering are starting to provide solutions for high-throughput imaging [58•].

Micropatterning approaches allow quantitative assays of signaling interactions by promoting the 2D geometric confinement of cells cultured on micrometer patches of adhesive protein (collagen, fibronectin, lamin, and Matrigel). Micropatterns of defined geometries are generated by microcontact printing or stamping

Figure 3



(a) Geometry modulates morphogen-gradient establishment and interpretation in 2D micropatterns (top). Peripheral cells receive disproportionate signals due to lateral receptor localization (termed edge-sensing), and micropatterns self-organize morphogen gradients via an activator–inhibitor reaction–diffusion circuit. Lumen fusion in peri-implantation mouse embryogenesis promotes a topological transformation, establishing an epithelial 'edge' in the proximal epiblast, potentially allowing for analogous mechanisms to explain patterning *in vivo* (bottom). **(b)** Global coupling between patterning and morphogenesis: the case of 'pescoids'. Breaking symmetry to establish a focus of Nodal activity is required for pescoid axial elongation, and in turn, this elongation is required to generate separated domains of BMP and Wnt signaling. **(c)** Mechanosensation couples physical cues to cell-fate transitions: the case of E-cadherin and β-catenin. β-catenin can occupy two cellular locations: (i) associated with E-cadherin constituting the adherens complex; and (ii) nuclear, promoting TCF-mediated transcription. Mechanical stimuli or a lack of E-cadherin-presenting neighbors (i.e. EMT) can promote transitions from the former to the latter, executing cell-fate changes. **(d)** Decoupling patterning and morphogenesis in a paradigm of mechanochemical coupling: somite morphogenesis in 3D gastruloids. Literature model of somitogenesis (top): rostro-caudal kinematic waves of Notch activity interact with the determination-front (the zone of low Fgf8) to establish somite boundaries (via Eph–Ephrin signaling) and polarity (marked by Tbx18 vs. Uncx4.1). 3D gastruloids embedded in Matrigel show clock-and-wavefront (CW) signatures and somite morphogenesis (bottom). Yet somites can form without a clock-provided alternate chemical stimuli, and molecular hallmarks of CW are manifest in the absence of somite morphogenesis (seen in gastruloids not embedded in Matrigel).

[22••,59–61]. Such studies have highlighted roles for cell density, colony size, and differential mechanics in modulating cellular responses to external chemical cues [50,59,62].

In vivo, fate acquisition and morphogenesis occur in the context of the extracellular matrix (ECM), which provides both mechanochemical cues and physical constraints. Hydrogel-based platforms offer bioengineered surrogates that promote appropriate transitions in fate and form. For example, ETS embryos rely on Matrigel to provide both mechanical constraint and biochemical cell polarization cues, necessary for the proamniotic cavity to open up [34••]. Intriguingly, Matrigel also promotes the generation of somite-like epithelial structures in gastruloids [63–65]. However, the impacts of the physical and chemical properties of various hydrogels on self-organization remain poorly defined.

Stem-cell microenvironments *in vivo* are dynamic and heterogeneous, in stark contrast to the inherently static and homogeneous nature of most culture platforms. Advances in the manufacture of microfluidic devices are overcoming this constraint, allowing stem cells to be cultured under precisely defined mechanochemical conditions, including the controlled provision of morphogen gradients. The new-found precision and robustness in patterning conferred by these technologies is illustrated in a model of the human amniotic sac, where a microfluidic device provides a polarized BMP4 signal that induces asymmetric morphological changes and gastrulation [66••].

What have we learned about natural embryogenesis using its synthetic counterpart?

Embryogenesis involves a complex interplay between signaling and morphogenesis. In spite of this, each process is often studied in isolation. With their amenability to manipulation, embryoids thus provide an opportunity to study this interplay, able to decouple processes canalized *in vivo*. Moreover, embryoids enable a bottom-up approach, providing the opportunity to apply physical, biochemical, and genetic methods to investigate mechanisms and downstream studies at high throughput.

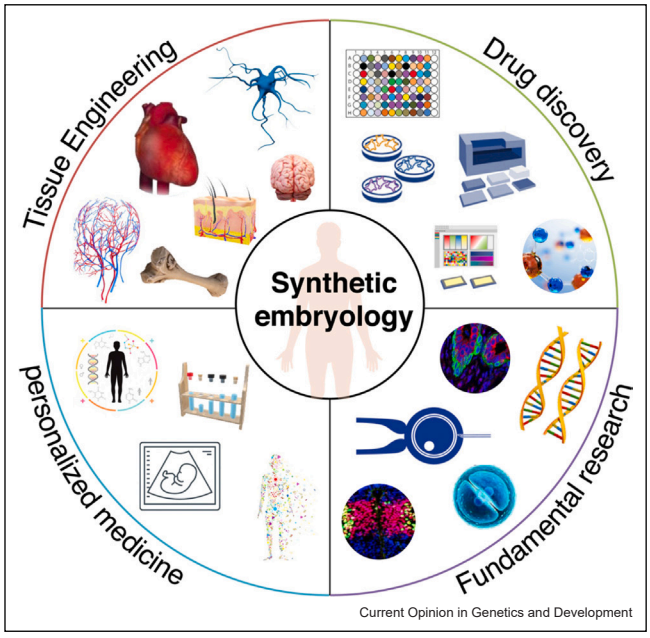
Morphogen gradients play critical roles in establishing spatially organized cell fates [67], yet less is known about how these chemical cues instruct changes in embryonic architecture. A combination of embryological and synthetic approaches has emphasized a role for mechanochemical feedback in mouse gastrulation initiation [68]. The BM, lining the epiblast–visceral endoderm interface, was found to display perforations that become biased toward the future posterior concomitant with anterior–posterior axis specification. Using ESC aggregates embedded in gelatin to model epiblast–BM interactions, perforations were shown to be orchestrated by posterior-biased Nodal-dependent expression of matrix metalloproteinases (MMP). Perforations have drastic consequences on morphogenesis, alleviating mechanical constraint from the BM: without perforations, ESC growth is reduced; and blocking MMP activity disrupts mesendodermal ingression and primitive streak extension.

Much like how morphogen gradients can shape morphogenesis, changes in embryonic form can have reciprocal impacts on patterning. Geometric influences are highlighted in the aforementioned studies of germ-layer patterning in 2D micropatterned hESCs [22••,50,69,70]. Patterning through the BMP pathway is in part achieved by edge-sensing (Figure 3a): while central cells can only receive signals basolaterally owing to polarized receptor localization, peripheral cells are biased in their signal reception through a more distributed receptor localization across the plasma membrane [50]. Gradients of BMP activity are established by a reaction–diffusion mechanism, with the extracellular BMP antagonist Noggin accruing at highest concentrations centrally [50]. Hence, embryo geometry — affecting morphogen diffusion — and topology — in the form of epithelial 'edges' — can influence patterning.

In some cases, morphogenesis and patterning are mutually dependent. This may occur at a global scale, where gradients shape mechanical cell behaviors, and morphogenesis shapes gradient, as seen in zebrafish embryoids — 'pescoids' — generated by reaggregating early-embryonic explants [71]. Pescoids break symmetry specifying mesendoderm and undergo axial elongation (Figure 3b). Nodal gradients established in symmetry breaking are necessary for elongation, promoting planar-cell-polarity signaling, which elicits convergent extension [71,72]. Conversely, elongation is necessary for axial specification.

Mutual dependence may also occur at the cell scale where changes in fate and physical properties are coupled via mechanotransduction. Studies in Drosophila and zebrafish suggest that mechanosensation may mediate mesoderm induction [73]: mechanical stimuli promote β-catenin release from adherens junctions to facilitate nuclear translocation (Figure 3c). Similar principles may hold in mammalian synthetic models. Patterns of anisotropic tension at colony edges are observed in hESC micropatterns of alternate shapes, with mesoderm induction biased toward foci of high stress [74]. Moreover, peripherally biased E-cadherin localization in circular micropatterns is necessary for the patterning of Wnt signaling, with a peripheral-to-central EMT wave proposed to promote sequential activation of the Wnt

Figure 4



Potential applications of synthetic embryology. Synthetic embryology should allow the recapitulation of different human organs/tissues, and should be amenable to genetic modifications, personalized medicine approaches, cancer studies, modeling of disease, drug discovery, high-throughput screenings, and fundamental research.

pathway via E-cadherin destabilization and β-catenin nuclear translocation [70].

Somitogenesis is perhaps the most paradigmatic example of the coupling of mechanics and patterning. Commencing after gastrulation, bilateral streaks of presomitic mesoderm undergo sequential budding in a rostral–caudal fashion, establishing epithelial rosettes — somites [75]. Somitogenesis has classically been explained by the 'clock and wavefront model' (C&W) [76–78]. Culture of gastruloids has unveiled molecular signatures of both clock and wavefront dynamics [63,64]. Yet, gastruloids lack somite morphology, suggesting patterning in the absence of morphogenesis. This contrasts with the embryo, where Notch and Eph–Ephrin signaling specifies epithelial boundaries [79]. Somite-like structures, however, can be established, provided aggregates are embedded in Matrigel [63–65]. Indeed, excessive Wnt signaling combined with BMP inhibition can promote a 'bunch of grapes' organization of somites, arranged randomly with respect to the C&W axis [80] (Figure 3d). However, the observed epithelial structures might also be due to a self-organizing pattern of differentiating paraxial mesoderm derivatives, as is observed for neural rosettes in differentiating ESCs [81]. Thus, given the separability of patterning and morphogenesis, synthetic models can interrogate somite morphogenesis, especially with regard to the role of the external environment in modulating cell mechanics, polarity, and state.

Conclusion, challenges, and future perspectives

Synthetic embryology aims to use ESCs to recapitulate key stages of embryo development, including morphogenesis, body-axis establishment, and cell-fate assignment. We predict that synthetic embryology will provide important insights into the key interactions and feedback mechanisms regulating cell/tissue morphogenesis and cell-fate specification during embryogenesis.

Most *in vitro* strategies to generate embryo-like structures from stem cells yield highly heterogeneous populations, varying in gene-expression profiles and/or morphology. While these differences may provide clues about the self-organization mechanisms underpinning synthetic and natural embryogenesis, future protocols should aspire to canalize their development in order to generate reliable experimental models with translational potential. To this end, we argue that experimental design must consider three parameters: (i) which cells? Certain stem-cell lines are notoriously heterogeneous, likely eliciting unwanted biochemical and mechanical impacts on self-organization, and compromise the robustness and reproducibility of the approach. It should be noted that the developmental potentials of most stem-cell types are lineage-restricted, so that they have difficulties in crossing other lineage boundaries, totipotent embryonic stem cells (TESCs) hold the greatest developmental potency, and recent progress in

recapitulating TESCs *in vitro* with unrestricted potency would have applications for synthetic embryology [82–85••]. (ii) Which chemical environment? Mechanochemical interactions between stem-cell types remain underexplored. These interactions complicate explanations of morphogenesis in multi-stem-cell models such as embryoids and blastoids, and necessitate a trade-off to be struck between exogenous versus endogenous supply of necessary growth factors and morphogens. Solving this problem requires a deeper understanding of molecular networks and their impact on cellular dynamics. (iii) Which bioengineered platform? Feasibly the most essential component of all is the mechanical environment, with bioengineered platforms becoming a fundamental constraint on self-assembly. Therefore, in order to improve the efficiency of stem-cell-based embryos, a deep understanding of each network components (cell, biochemical, and biophysical environment) is crucial. Fortunately, modern tools are helping us to control the fate and form of synthetic models with increased precision. Previous work on engineering the development of organoids by precisely manipulating microenvironmental signals has yielded some insights into the morphogenetic rules and design parameters that promote self-organization [86]. Moreover, theoretical and computational modeling will become increasingly essential in predicting self-organization outcomes.

Another challenge of synthetic embryology field is the long-term culture of synthetic embryos. Supplying an optimized concentration of oxygen and suitable nutrients with perfusion should solve this problem. Moreover, engineering microenvironments for stem cells with controlled ECM components, suitable biophysical properties, and spatiotemporally tailored biochemical signals would facilitate enhanced developmental complexity, tuning fate and form down increasingly embryo-like routes.

Comparing the *in vitro* self-organization results with actual embryos is also a challenge, particularly for human models. Recently, some platforms have been developed for culturing postimplantation human and monkey embryos [87–89]. However, it is unclear to what extent human embryoids will mimic morphogenesis *in vivo*.

Overall, stem-cell-based models of mammalian embryogenesis are particularly well-suited to advance understanding of the molecular mechanisms and transcriptional networks that underpin embryo development. We envision embryo models to become a mainstay of research into human development, dovetailing with increasingly detailed molecular elucidation of embryogenesis in our own species. Synthetic embryology also holds great potential to be a spearhead for interdisciplinary science, necessitating collaboration between researchers advancing single-cell assays, 3D cell printing, multi-omics, and optogenetics. Combining synthetic embryology with bioengineering will enable new breakthroughs in tissue engineering, drug discovery, personalized medicine, regenerative therapeutics, and fundamental research (Figure 4).

Conflict of interest statement

The authors declare no competing interests.

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- • of special interest
- •• of outstanding interest

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